



## COURSE DESCRIPTION FORM

**INSTITUTION** National University of Computer and Emerging Sciences (NUCES-FAST)  
BS(CS), BS(CY), BS(SE), BS(AI)

**PROGRAM (S) TO BE  
EVALUATED**

### A. Course Description

|                                                                                                                             |                                                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Course Code</b>                                                                                                          | CS-1004                                                                                                                                                                                                                                    |
| <b>Course Title</b>                                                                                                         | Object-oriented Programming                                                                                                                                                                                                                |
| <b>Credit Hours</b>                                                                                                         | 3+1                                                                                                                                                                                                                                        |
| <b>Prerequisites by Course(s) and Topics</b>                                                                                | Programming Fundamentals (CS-1002)                                                                                                                                                                                                         |
| <b>Assessment Instruments with Weights</b><br>(homework, quizzes, midterms, final, programming assignments, lab work, etc.) | <b><u>Theory:</u></b><br>Mid-1: 15<br>Mid-2: 15<br>Quizzes: 12 (4 total)<br>Assignments: 8 (2 total : 4+4)<br>Final: 50<br><br><b><u>Lab:</u></b><br>Lab Activities: 20 (2 each and best 10)<br>Midterm : 20<br>Project : 10<br>Final : 50 |



**UNIVERSITY OF  
SALAMANCA**

| <b>Lab Coordinator</b>                                          | Muhammad Khalid                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                 |  |  |    |                                                                                                                                                   |  |    |                                                                                      |  |    |                                                                                                                      |  |    |                                                                                                                       |  |    |                                                                                    |  |                              |  |  |                        |                                                                                                                                                         |   |  |  |   |
|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|--|--|----|---------------------------------------------------------------------------------------------------------------------------------------------------|--|----|--------------------------------------------------------------------------------------|--|----|----------------------------------------------------------------------------------------------------------------------|--|----|-----------------------------------------------------------------------------------------------------------------------|--|----|------------------------------------------------------------------------------------|--|------------------------------|--|--|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---|--|--|---|
| <b>URL (if any)</b>                                             | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                 |  |  |    |                                                                                                                                                   |  |    |                                                                                      |  |    |                                                                                                                      |  |    |                                                                                                                       |  |    |                                                                                    |  |                              |  |  |                        |                                                                                                                                                         |   |  |  |   |
| <b>Current Catalog Description</b>                              | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                 |  |  |    |                                                                                                                                                   |  |    |                                                                                      |  |    |                                                                                                                      |  |    |                                                                                                                       |  |    |                                                                                    |  |                              |  |  |                        |                                                                                                                                                         |   |  |  |   |
| <b>Textbook (or Laboratory Manual for Laboratory Courses)</b>   | <p><b><u>Textbook:</u></b></p> <ol style="list-style-type: none"> <li>1. "Problem Solving with C++", 9e Global Edition, Walter Savitch, ISBN-13:9781292018249, Addison-Wesley, 2015.</li> <li>2. C++ How to program By Deitel &amp; Deitel.</li> </ol> <p><b><u>Reference books:</u></b></p> <ol style="list-style-type: none"> <li>1. The C++ Programming Language by Bjarne Stroustrup.</li> <li>2. Object Oriented Software Engineering by Jacobson.</li> <li>3. C# 4.0: The Complete Reference by Herbert Schildt</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                 |  |  |    |                                                                                                                                                   |  |    |                                                                                      |  |    |                                                                                                                      |  |    |                                                                                                                       |  |    |                                                                                    |  |                              |  |  |                        |                                                                                                                                                         |   |  |  |   |
| <b>Reference Material</b>                                       | GCR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                 |  |  |    |                                                                                                                                                   |  |    |                                                                                      |  |    |                                                                                                                      |  |    |                                                                                                                       |  |    |                                                                                    |  |                              |  |  |                        |                                                                                                                                                         |   |  |  |   |
| <b>Course Goals</b>                                             | <table border="1"> <tr> <th align="left" colspan="3">A. Course Learning Outcomes (CLOs) with Bloom's Taxonomy Levels</th> </tr> <tr> <td>1.</td><td><b>Discuss</b> knowledge of underlying concepts of object-oriented paradigm like abstraction, encapsulation, polymorphism, inheritance etc. (C-2)</td><td></td> </tr> <tr> <td>2.</td><td><b>Identify</b> real world problems in terms of objects rather than procedure. (C-4)</td><td></td> </tr> <tr> <td>3.</td><td><b>Illustrate</b> Object-Oriented design artifacts and their mapping to Object-Oriented Programming using C++. (C-3)</td><td></td> </tr> <tr> <td>4.</td><td><b>Design</b> and assess small and medium scale C++ / C# programs using object-oriented programming principles. (C-6)</td><td></td> </tr> <tr> <td>5.</td><td><b>Synthesize</b> programs using Generic Programming and exception handling. (C-6)</td><td></td> </tr> <tr> <th align="left" colspan="3">B. Program Learning Outcomes</th> </tr> <tr> <td>1. Computing Knowledge</td><td>Apply knowledge of mathematics, natural sciences, computing fundamentals, and a computing specialization to the solution of complex computing problems.</td><td align="center">✓</td> </tr> <tr> <td></td><td></td><td align="center">✓</td> </tr> </table> | A. Course Learning Outcomes (CLOs) with Bloom's Taxonomy Levels |  |  | 1. | <b>Discuss</b> knowledge of underlying concepts of object-oriented paradigm like abstraction, encapsulation, polymorphism, inheritance etc. (C-2) |  | 2. | <b>Identify</b> real world problems in terms of objects rather than procedure. (C-4) |  | 3. | <b>Illustrate</b> Object-Oriented design artifacts and their mapping to Object-Oriented Programming using C++. (C-3) |  | 4. | <b>Design</b> and assess small and medium scale C++ / C# programs using object-oriented programming principles. (C-6) |  | 5. | <b>Synthesize</b> programs using Generic Programming and exception handling. (C-6) |  | B. Program Learning Outcomes |  |  | 1. Computing Knowledge | Apply knowledge of mathematics, natural sciences, computing fundamentals, and a computing specialization to the solution of complex computing problems. | ✓ |  |  | ✓ |
| A. Course Learning Outcomes (CLOs) with Bloom's Taxonomy Levels |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                 |  |  |    |                                                                                                                                                   |  |    |                                                                                      |  |    |                                                                                                                      |  |    |                                                                                                                       |  |    |                                                                                    |  |                              |  |  |                        |                                                                                                                                                         |   |  |  |   |
| 1.                                                              | <b>Discuss</b> knowledge of underlying concepts of object-oriented paradigm like abstraction, encapsulation, polymorphism, inheritance etc. (C-2)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                 |  |  |    |                                                                                                                                                   |  |    |                                                                                      |  |    |                                                                                                                      |  |    |                                                                                                                       |  |    |                                                                                    |  |                              |  |  |                        |                                                                                                                                                         |   |  |  |   |
| 2.                                                              | <b>Identify</b> real world problems in terms of objects rather than procedure. (C-4)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                 |  |  |    |                                                                                                                                                   |  |    |                                                                                      |  |    |                                                                                                                      |  |    |                                                                                                                       |  |    |                                                                                    |  |                              |  |  |                        |                                                                                                                                                         |   |  |  |   |
| 3.                                                              | <b>Illustrate</b> Object-Oriented design artifacts and their mapping to Object-Oriented Programming using C++. (C-3)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                 |  |  |    |                                                                                                                                                   |  |    |                                                                                      |  |    |                                                                                                                      |  |    |                                                                                                                       |  |    |                                                                                    |  |                              |  |  |                        |                                                                                                                                                         |   |  |  |   |
| 4.                                                              | <b>Design</b> and assess small and medium scale C++ / C# programs using object-oriented programming principles. (C-6)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                 |  |  |    |                                                                                                                                                   |  |    |                                                                                      |  |    |                                                                                                                      |  |    |                                                                                                                       |  |    |                                                                                    |  |                              |  |  |                        |                                                                                                                                                         |   |  |  |   |
| 5.                                                              | <b>Synthesize</b> programs using Generic Programming and exception handling. (C-6)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                 |  |  |    |                                                                                                                                                   |  |    |                                                                                      |  |    |                                                                                                                      |  |    |                                                                                                                       |  |    |                                                                                    |  |                              |  |  |                        |                                                                                                                                                         |   |  |  |   |
| B. Program Learning Outcomes                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                 |  |  |    |                                                                                                                                                   |  |    |                                                                                      |  |    |                                                                                                                      |  |    |                                                                                                                       |  |    |                                                                                    |  |                              |  |  |                        |                                                                                                                                                         |   |  |  |   |
| 1. Computing Knowledge                                          | Apply knowledge of mathematics, natural sciences, computing fundamentals, and a computing specialization to the solution of complex computing problems.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | ✓                                                               |  |  |    |                                                                                                                                                   |  |    |                                                                                      |  |    |                                                                                                                      |  |    |                                                                                                                       |  |    |                                                                                    |  |                              |  |  |                        |                                                                                                                                                         |   |  |  |   |
|                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | ✓                                                               |  |  |    |                                                                                                                                                   |  |    |                                                                                      |  |    |                                                                                                                      |  |    |                                                                                                                       |  |    |                                                                                    |  |                              |  |  |                        |                                                                                                                                                         |   |  |  |   |



|  |                                    |                                                                                                                                                                                                                                            |   |
|--|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|  | 2. Problem Analysis                | Identify, formulate, research literature, and analyze complex computing problems, reaching substantiated conclusions using first principles of mathematics, natural sciences, and computing sciences.                                      |   |
|  | 3.Design/Develop Solutions         | Design solutions for complex computing problems and design systems, components, and processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations. | ✓ |
|  | 4. Investigation & Experimentation | Conduct investigation of complex computing problems using research-based knowledge and research-based methods                                                                                                                              |   |
|  | 5. Modern Tool Usage               | Create, select, and apply appropriate techniques, resources and modern computing tools, including prediction and modelling for complex computing problems.                                                                                 |   |
|  | 6. Society Responsibility          | Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal, and cultural issues relevant to context of complex computing problems.                                                                         |   |
|  | 7. Environment and Sustainability  | Understand and evaluate sustainability and impact of professional computing work in the solution of complex computing problems                                                                                                             |   |
|  | 8. Ethics                          | Apply ethical principles and commit to professional ethics and responsibilities and norms of computing practice                                                                                                                            |   |
|  |                                    |                                                                                                                                                                                                                                            |   |



|                                                                                                                                                                                                                                                                             | 9. Individual and Team Work                                                                                 | Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.                                          |           |            |   |   |   |   |   |   |    |    |    |      |       |     |           |            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------|---|---|---|---|---|---|----|----|----|------|-------|-----|-----------|------------|
|                                                                                                                                                                                                                                                                             | 10. Communication                                                                                           | Communicate effectively on complex computing activities with the computing community and with society at large.                                                |           |            |   |   |   |   |   |   |    |    |    |      |       |     |           |            |
|                                                                                                                                                                                                                                                                             | 11. Project Management and Finance                                                                          | Demonstrate knowledge and understanding of management principles and economic decision making and apply these to one's own work as a member or a team.         |           |            |   |   |   |   |   |   |    |    |    |      |       |     |           |            |
|                                                                                                                                                                                                                                                                             | 12. Life Long Learning                                                                                      | Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological changes. |           |            |   |   |   |   |   |   |    |    |    |      |       |     |           |            |
|                                                                                                                                                                                                                                                                             | <b>C. Relation between CLOs and PLOs</b><br>(CLO: Course Learning Outcome, PLOs: Program Learning Outcomes) |                                                                                                                                                                |           |            |   |   |   |   |   |   |    |    |    |      |       |     |           |            |
|                                                                                                                                                                                                                                                                             |                                                                                                             | <b>PLOs</b>                                                                                                                                                    |           |            |   |   |   |   |   |   |    |    |    |      |       |     |           |            |
|                                                                                                                                                                                                                                                                             |                                                                                                             | 1                                                                                                                                                              | 2         | 3          | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |      |       |     |           |            |
| <b>C<br/>L<br/>O<br/>s</b>                                                                                                                                                                                                                                                  | 1                                                                                                           | ✓                                                                                                                                                              |           |            |   |   |   |   |   |   |    |    |    |      |       |     |           |            |
|                                                                                                                                                                                                                                                                             | 2                                                                                                           | ✓                                                                                                                                                              |           |            |   |   |   |   |   |   |    |    |    |      |       |     |           |            |
|                                                                                                                                                                                                                                                                             | 3                                                                                                           |                                                                                                                                                                | ✓         |            |   |   |   |   |   |   |    |    |    |      |       |     |           |            |
|                                                                                                                                                                                                                                                                             | 4                                                                                                           |                                                                                                                                                                |           | ✓          |   |   |   |   |   |   |    |    |    |      |       |     |           |            |
|                                                                                                                                                                                                                                                                             | 5                                                                                                           |                                                                                                                                                                | ✓         |            |   |   |   |   |   |   |    |    |    |      |       |     |           |            |
| <table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <th style="width: 10%;">Week</th> <th style="width: 30%;">Topic</th> <th style="width: 10%;">CLO</th> <th style="width: 30%;">Lab Topic</th> <th style="width: 20%;">Assessment</th> </tr> </table> |                                                                                                             |                                                                                                                                                                |           |            |   |   |   |   |   |   |    |    |    | Week | Topic | CLO | Lab Topic | Assessment |
| Week                                                                                                                                                                                                                                                                        | Topic                                                                                                       | CLO                                                                                                                                                            | Lab Topic | Assessment |   |   |   |   |   |   |    |    |    |      |       |     |           |            |
| <b>Topics Covered in the Course, with Number of Lectures</b>                                                                                                                                                                                                                |                                                                                                             |                                                                                                                                                                |           |            |   |   |   |   |   |   |    |    |    |      |       |     |           |            |



|                                                                            |    |                                                        |         |                                                                                            |                                                 |
|----------------------------------------------------------------------------|----|--------------------------------------------------------|---------|--------------------------------------------------------------------------------------------|-------------------------------------------------|
| <b>on Each Topic</b><br>(assume 15-week instruction and one-hour lectures) | 1  | Introduction to OO paradigm                            | 1       | Introduction to IDE, skeleton of C++ program, double pointers, 2d arrays, basic I/O in C++ | Assignment 1<br>Quiz 1 Week 3                   |
|                                                                            |    | Comparison from sequential & procedural paradigms      | 1       | C++ data types, functions, struct revisited based on real world use cases                  |                                                 |
|                                                                            |    | Data Abstraction Encapsulation                         | 1       |                                                                                            |                                                 |
|                                                                            | 2  | Introduction to Objects in real world                  | 1,2     | Classes & Objects                                                                          |                                                 |
|                                                                            |    | Introduction to classes and objects                    | 1,2,3   |                                                                                            |                                                 |
|                                                                            |    | Access Control                                         | 1,2,3   |                                                                                            |                                                 |
|                                                                            |    | Constructors & its types, Destructor                   | 1,3,4   |                                                                                            |                                                 |
|                                                                            | 3  | Setters & Getters                                      | 1,3,4   | Working with classes and Constructors, setters and getters, Working with access modifiers  |                                                 |
|                                                                            |    | Member initialization list                             | 1,3     |                                                                                            |                                                 |
|                                                                            |    | Constants, Constants with pointers, constant functions | 1,3     |                                                                                            |                                                 |
|                                                                            | 4  | Static data and member functions,                      | 1,3     | Static and constant keywords, This pointer                                                 |                                                 |
|                                                                            |    | Inline functions, This pointer                         | 1,3     |                                                                                            |                                                 |
|                                                                            | 5  | Array of objects Has-a relation                        | 1,2,3,4 | Array of objects Has-a relation                                                            |                                                 |
|                                                                            | 6  | <b>Mid I Exam</b>                                      |         |                                                                                            |                                                 |
|                                                                            | 7  | Introduction of Inheritance                            | 1,2,3,4 | Member initialization list Inheritance                                                     | Assignment 2<br>Quiz 2 Week 7<br>Quiz 3 Week 10 |
|                                                                            |    | Types of inheritance                                   |         |                                                                                            |                                                 |
|                                                                            |    | Data and code hiding                                   | 1,2,3,4 |                                                                                            |                                                 |
|                                                                            | 8  | Polymorphism in OOP                                    | 1,2,3,4 | Polymorphism, Function overloading and overriding                                          |                                                 |
|                                                                            |    | Function overriding and overloading                    | 1,2,3,4 |                                                                                            |                                                 |
|                                                                            |    |                                                        | 1,2,3,4 |                                                                                            |                                                 |
|                                                                            | 9  | Friend function                                        | 1,2,3,4 | Lab Mid                                                                                    |                                                 |
|                                                                            |    | Operator overloading                                   | 1,2,3,4 |                                                                                            |                                                 |
|                                                                            | 10 | Multiple inheritance & its issues (Diamond Problem)    | 1,2,3,4 | Friend classes, Friend functions, operator overloading                                     |                                                 |
|                                                                            |    | Virtual inheritance                                    | 1,2,3,4 |                                                                                            |                                                 |
|                                                                            |    | Virtual functions                                      | 1,2,3,4 |                                                                                            |                                                 |
|                                                                            | 11 | Abstract classes & Interfaces                          | 1,2,3,4 | Abstract Classes and virtual functions                                                     |                                                 |
|                                                                            | 12 | <b>Mid II Exam</b>                                     |         |                                                                                            |                                                 |
|                                                                            | 13 |                                                        | 1,2,3,4 | Filing and I/O stream                                                                      |                                                 |



**FORM 001**

|                                                    |                                                                                                                                                                                                                                                                                                               |                                    |                 |                                                      |        |
|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|-----------------|------------------------------------------------------|--------|
|                                                    |                                                                                                                                                                                                                                                                                                               | Introduction to filing             |                 |                                                      |        |
|                                                    | 14                                                                                                                                                                                                                                                                                                            | Filing Continue Generics           | 5               | Working with template functions and template classes | Quiz 4 |
|                                                    |                                                                                                                                                                                                                                                                                                               | Generics and Templates             | 5               |                                                      |        |
|                                                    | 15                                                                                                                                                                                                                                                                                                            | Introduction to exception handling | 1,2             | Project Submission & Project demo                    |        |
|                                                    |                                                                                                                                                                                                                                                                                                               | STL (Vector, List)                 | 1,2             |                                                      |        |
|                                                    | 16                                                                                                                                                                                                                                                                                                            | Final lab exam                     |                 |                                                      |        |
|                                                    |                                                                                                                                                                                                                                                                                                               | Final Exam                         |                 |                                                      |        |
| Laboratory Projects/Experiments Done in the Course | 1                                                                                                                                                                                                                                                                                                             |                                    |                 |                                                      |        |
| Programming Assignments Done in the Course         | 3 Assignments                                                                                                                                                                                                                                                                                                 |                                    |                 |                                                      |        |
| Class Time Spent on (in credit hours)              | Theory                                                                                                                                                                                                                                                                                                        | Problem Analysis                   | Solution Design | Social and Ethical Issues                            |        |
|                                                    | 15                                                                                                                                                                                                                                                                                                            | 15                                 | 13              | 0                                                    |        |
| Oral and Written Communications                    | Every student is required to submit at least _1__written report of typically _2__pages and to make _1__oral presentations of typically _10_ minute’s duration. Include only material that is graded for grammar, spelling, style, and so forth, as well as for technical content, completeness, and accuracy. |                                    |                 |                                                      |        |

**Instructor Name:**

Izzah Salam

**Date:**

15th January, 2026